

IMPLEMENT SHAP AND LIME USING AN ML MODEL FOR CHILD HEALTH IN NEONATAL CARE

Explainable Machine Learning Project

Submitted project package

GitHub repository: https://github.com/YOUR_USERNAME/neonatal-health-shap-lime

Note: The included modeling dataset is synthetic and intended only for academic demonstration. The system is not a medical diagnostic or clinical decision-support tool.

1. Abstract

This project demonstrates an explainable machine-learning workflow for neonatal health risk prediction. A Random Forest classifier is trained on a synthetic tabular dataset containing neonatal vital signs, Apgar scores, birth characteristics and maternal factors. SHAP (SHapley Additive exPlanations) is used to explain feature contributions at the global and individual levels. LIME (Local Interpretable Model-agnostic Explanations) is demonstrated through a local surrogate model that perturbs an observation, queries the black-box classifier, and learns a weighted linear approximation around that observation. The final package also contains a Streamlit interface for entering an example record and viewing the prediction with both explanation methods.

2. Problem Statement

Machine-learning predictions can be difficult to interpret, especially in healthcare-related applications. The objective of this project is to create a neonatal health risk prediction model and integrate explainability methods so that users can understand which input variables influenced a prediction. The project focuses on technical explainability rather than clinical diagnosis.

3. Objectives

- Build a supervised ML classification model for an educational neonatal-risk task.
- Preprocess numerical and categorical health-related features.
- Evaluate the model using accuracy, precision, recall, F1-score and ROC-AUC.
- Integrate SHAP for global and local model explanations.
- Integrate LIME/local surrogate explanations for individual predictions.
- Provide a Streamlit demonstration interface and reproducible GitHub-ready project.

4. Dataset

The repository contains 3,000 synthetic neonatal records. Variables include gestational age, birth weight, Apgar scores, maternal age, prenatal visits, delivery mode, sex, temperature, heart rate, respiratory rate and oxygen saturation. The target is a synthetic binary label named `high_risk`. The label is generated from a probabilistic educational rule so that the dataset can demonstrate classification and explainability without using private patient information.

WHO defines the neonatal period as the first 28 completed days of life and reports neonatal mortality rate as deaths per 1,000 live births. UNICEF provides downloadable neonatal mortality datasets. These sources provide domain context; they are not the source of the synthetic individual-level records included here.

5. System Workflow

Dataset → Data preprocessing → Train/test split → Random Forest → Prediction → SHAP explanation + LIME explanation → Streamlit visualization

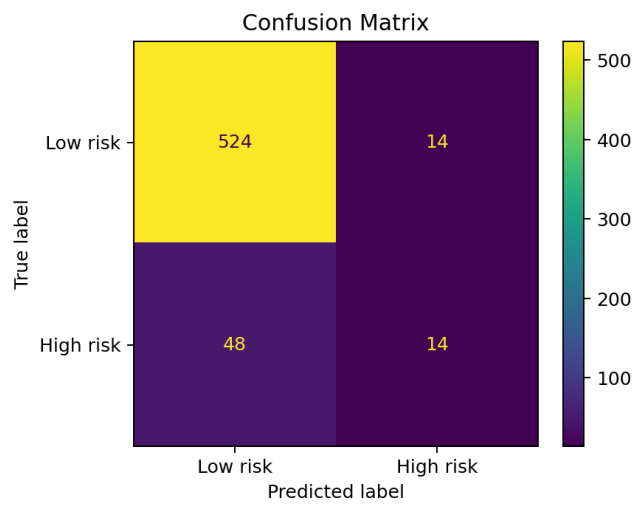
6. Machine Learning Method

A Random Forest classifier with 300 trees is used. Numerical variables are standardized and categorical variables are one-hot encoded using a `ColumnTransformer`. The preprocessing and classifier are stored together in a scikit-learn Pipeline, which reduces the risk of applying inconsistent preprocessing during prediction. The train/test split is stratified and uses an 80/20 ratio.

7. Model Performance

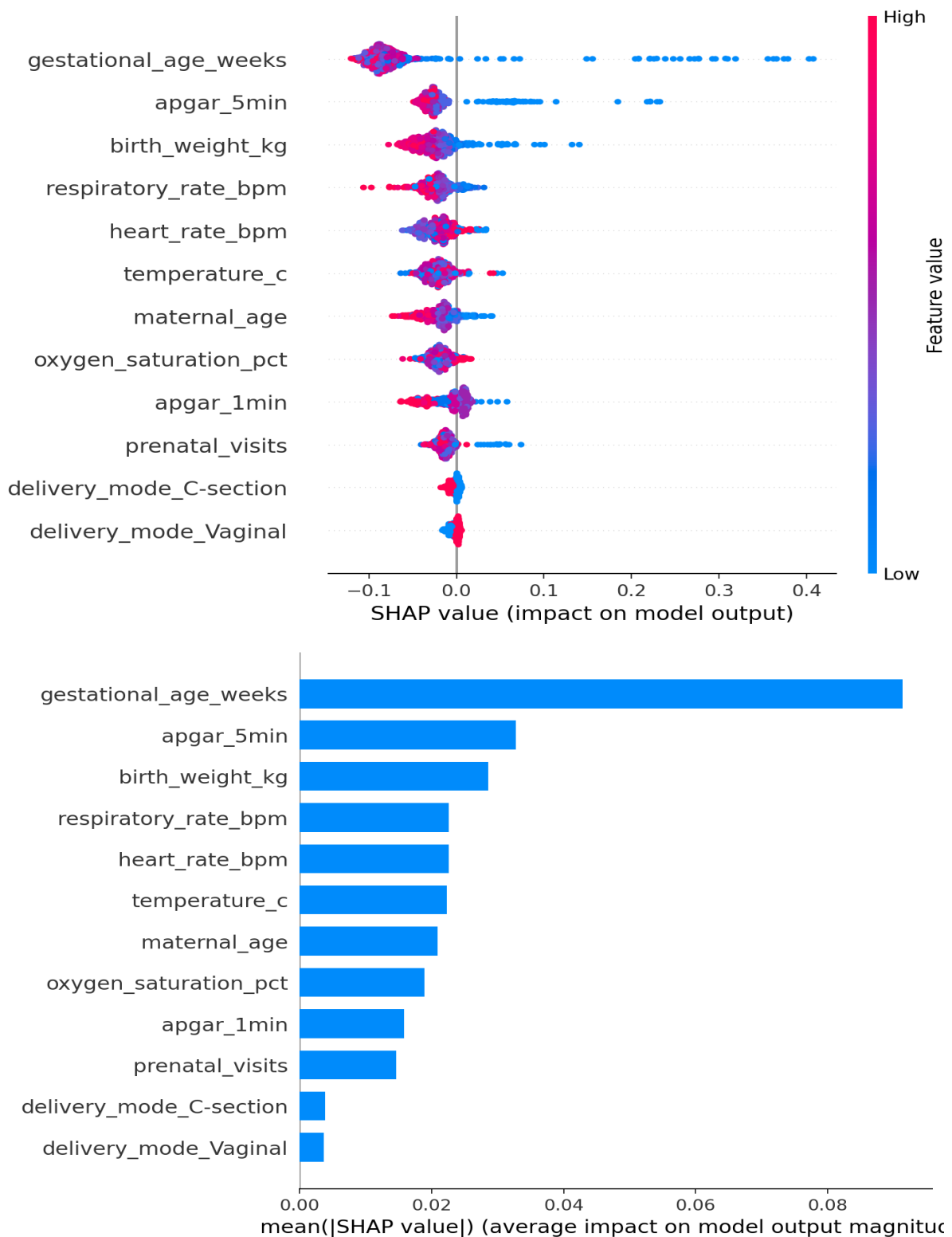
Metric	Value
Accuracy	0.8967
Precision	0.5000
Recall	0.2258
F1-score	0.3111
ROC-AUC	0.6651

The positive class is intentionally less frequent in the synthetic dataset, so accuracy alone should not be used to judge the model. Precision, recall, F1-score and ROC-AUC are included to provide a broader evaluation.



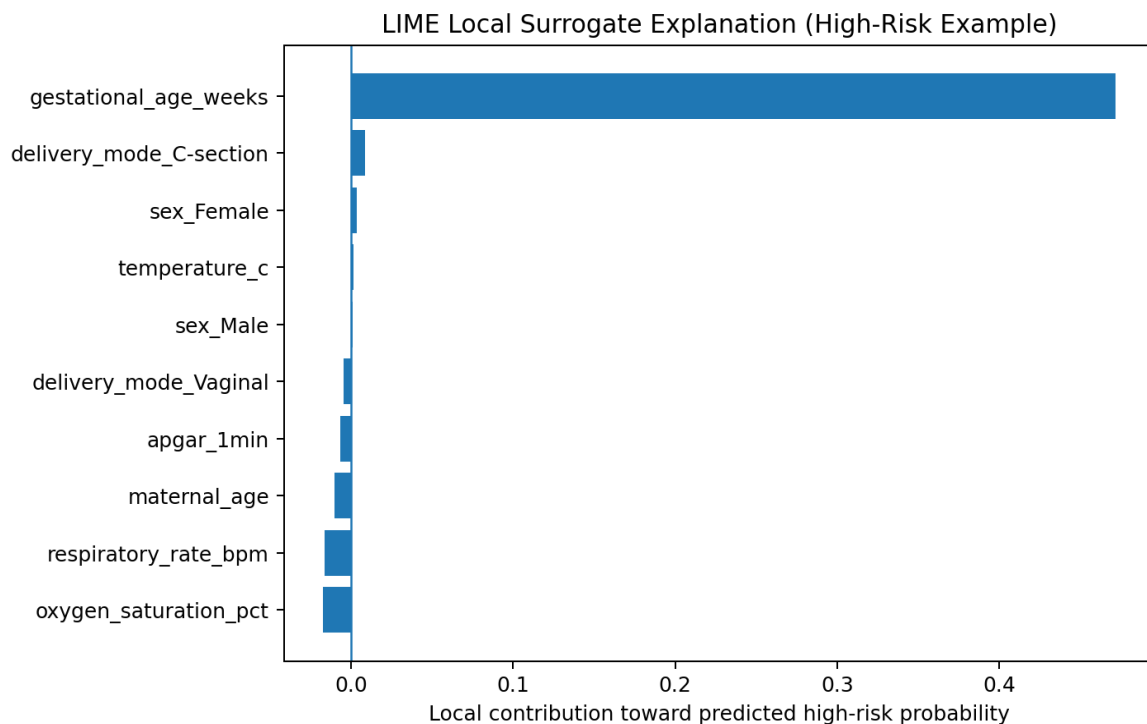
8. SHAP Implementation

SHAP explains predictions using Shapley-value-based feature contributions. TreeExplainer is applied to the trained Random Forest. The summary plot shows how strongly each transformed feature contributes across the evaluation sample, while the bar chart ranks average absolute contribution.



9. LIME Implementation

LIME explains one prediction locally by creating perturbed samples around the selected observation, obtaining black-box predictions for those samples, weighting nearby samples more strongly, and fitting a simple weighted linear surrogate. The project includes this implementation in `src/lime_local.py`. The requirements file also lists the official lime package so the official `LimeTabularExplainer` can be used when the package is installed.



The example above is a local explanation for a high-risk-probability test observation. Positive contributions move the local surrogate toward the high-risk class, while negative contributions move it away. The exact numerical contributions depend on the fitted model and perturbation settings.

10. SHAP vs LIME

Aspect	SHAP	LIME
Main idea	Shapley-value feature contributions	Local surrogate approximation
Scope	Global and local explanations	Primarily local explanations
Model dependence	TreeExplainer is optimized for tree models	Model-agnostic
Output	Feature contribution values and plots	Local feature weights
Use in this project	Random Forest global/local explanation	Individual prediction explanation

11. Streamlit Application

The Streamlit application in `app.py` provides input fields for the neonatal variables. When the user clicks Predict and explain, the application displays the predicted high-risk probability and tabular SHAP and LIME explanations. This creates a simple interface for demonstrating explainable AI without exposing a real patient record.

Run command:

```
streamlit run app.py
```

12. Project Structure

```
data/neonatal_health_synthetic.csv
src/train_model.py
src/lime_local.py
src/explain_model.py
src/neonatal_risk_model.joblib
app.py
requirements.txt
results/shap_summary.png
results/shap_feature_importance.png
results/lime_local_explanation.png
results/confusion_matrix.png
report/Neonatal_SHAP_LIME_Report.pdf
```

13. Conclusion

The project demonstrates an end-to-end explainable ML workflow for a neonatal health classification task. The Random Forest model provides the prediction layer, SHAP provides feature-attribution explanations, and the LIME-style local surrogate provides an interpretable approximation around an individual prediction. The architecture can be adapted to an approved real neonatal dataset by replacing the synthetic CSV and retraining the pipeline.

14. Future Enhancements

- Replace the synthetic dataset with an approved, de-identified neonatal dataset.
- Address class imbalance using clinically justified sampling, weighting or threshold selection.
- Compare Random Forest with gradient boosting and calibrated models.
- Add cross-validation and confidence/calibration analysis.
- Use official LIME Tabular Explainer in the deployment environment.
- Add model monitoring, fairness checks and dataset/version tracking.
- Validate the model prospectively before considering any real-world application.

15. GitHub Submission

GitHub repository to create: https://github.com/YOUR_USERNAME/neonatal-health-shap-lime

After creating the repository, upload the complete project folder and replace `YOUR_USERNAME` in this PDF/README with the actual GitHub username. The project includes a `.gitignore` and the commands required to push the code.

16. References

1. World Health Organization. Neonatal mortality rate indicator and metadata.
<https://www.who.int/data/gho/data/indicators/indicator-details/GHO/neonatal-mortality-rate-%28per-1000-live-births%29>
2. UNICEF Data. Neonatal mortality data. <https://data.unicef.org/resources/dataset/neonatal-mortality-data/>
3. World Health Organization. Child deaths in neonates.
<https://www.who.int/data/gho/data/indicators/indicator-details/GHO/number-of-neonatal-deaths>
4. Lundberg, S. M. & Lee, S.-I. A Unified Approach to Interpreting Model Predictions. NeurIPS, 2017.
5. Ribeiro, M. T., Singh, S. & Guestrin, C. Why Should I Trust You? Explaining the Predictions of Any Classifier. KDD, 2016.